LAB 7

Contrastive / Metric Learning on MagnaTagATune (MTAT)

Assignment objectives

1. Train a metric learning model on the MagnaTagATune (MTAT) dataset using contrastive learning (triplet loss and InfoNCE).
2. Evaluate the embedding quality:
    - Visualize embeddings in 2D for two non-overlapping tag groups (multi-label dataset).
    - Provide separate plots for each group.
3. Write a short report (in this notebook) describing decisions and observations.

Notebook steps (lecture alignment)

1. Train a CNN with triplet loss
2. Plot validation embeddings and clustering results
3. Train a CNN with InfoNCE loss
4. Plot validation embeddings and clustering results

Extras

- runnable on Modal, Colab, and local MacBook
- resource usage + environmental impact tracking
- basic economic cost analysis for training/inference

SETUP

In [23]:
```python
import os, sys, shutil, subprocess
import torch
from pathlib import Path

print("python:", sys.version)
print("torch:", torch.__version__)

ff = shutil.which("ffmpeg")
print("ffmpeg in PATH:", ff)

if ff is None and ("google.colab" in sys.modules):
    subprocess.check_call(["apt-get", "update", "-y"])
    subprocess.check_call(["apt-get", "install", "-y", "ffmpeg"])
    ff = shutil.which("ffmpeg")
    print("ffmpeg in PATH (after install):", ff)

try:
    import torchaudio
    print("torchaudio:", torchaudio.__version__)
except Exception as e:
    print("torchaudio import error:", repr(e))

try:
    import torchcodec
    print("torchcodec:", getattr(torchcodec, "__version__", "unknown"))
except Exception as e:
    print("torchcodec import error:", repr(e))

if Path("/vol").is_dir():
    print("ls /vol:", os.listdir("/vol"))
    for d in Path("/vol").iterdir():
        if d.is_dir():
            try:
                print(str(d), "->", os.listdir(d)[:30])
            except Exception as e:
                print(str(d), "->", repr(e))

import os, sys, platform, subprocess

def pip_install(pkgs):
    subprocess.check_call([sys.executable, "-m", "pip", "install", "-q"] + pkgs)

pip_install([
    "torch", "torchaudio", "tqdm", "scikit-learn", "matplotlib",
    "umap-learn", "psutil", "codecarbon", "mirdata", "librosa", "soundfile"
])
```

```
python: 3.13.2 (main, Feb  4 2025, 14:51:09) [Clang 16.0.0 (clang-1600.0.26.6)]
torch: 2.9.1
ffmpeg in PATH: /opt/homebrew/bin/ffmpeg
torchaudio: 2.9.1
torchcodec: 0.9.0
```

In [24]:
```python
# suppress non-critical audio warnings
import warnings

warnings.filterwarnings(
    "ignore",
    message="PySoundFile failed.*"
)
warnings.filterwarnings(
    "ignore",
    message=".*audioread.*"
)
warnings.filterwarnings(
    "ignore",
    message="n_fft=.* is too large.*"
)
```

In [25]:
```python
# installs runtime deps
import sys, os, platform, subprocess

def pip_install(pkgs):
    subprocess.check_call([sys.executable, "-m", "pip", "install", "-q"] + pkgs)

pip_install([
    "torch", "torchaudio", "tqdm", "scikit-learn", "matplotlib",
    "umap-learn", "psutil", "codecarbon", "mirdata", "librosa"
])
```

Setup and reproducibility

This section configures:

- device selection (cpu / cuda / mps)
- random seeds
- filesystem paths (Modal / Colab / local)
- lightweight logging utilities (timing + resource tracking)

In [26]:
```python
# setup device and paths
import time, json, random, os, sys
import numpy as np
import torch

SEED = 42
random.seed(SEED)
np.random.seed(SEED)
torch.manual_seed(SEED)

def running_on_colab():
    return "google.colab" in sys.modules

def running_on_modal():
    return os.environ.get("MODAL_ENVIRONMENT", "") != ""

def pick_device():
    if torch.cuda.is_available():
        return "cuda"
    if getattr(torch.backends, "mps", None) and torch.backends.mps.is_available():
        return "mps"
    return "cpu"

DEVICE = pick_device()

# paths
if running_on_modal():
    DATA_ROOT = "/mnt/mtat-datalabs"
    RUNS_ROOT = "/vol/runs" if os.path.isdir("/vol/runs") else "/mnt/runs"
elif running_on_colab():
    DATA_ROOT = "/content/data/mtat"
    RUNS_ROOT = "/content/runs"
else:
    DATA_ROOT = "./data/mtat"
    RUNS_ROOT = "./runs"

os.makedirs(DATA_ROOT, exist_ok=True)
os.makedirs(RUNS_ROOT, exist_ok=True)

# environment summary
print("environment summary")
print("  running on colab:", running_on_colab())
print("  running on modal:", running_on_modal())
print("  torch.cuda.is_available:", torch.cuda.is_available())
print("  mps available:", getattr(torch.backends, "mps", None) is not None and torch.backends.mps.is_available())
print("  selected device:", DEVICE)
print("  data root:", DATA_ROOT)
print("  runs root:", RUNS_ROOT)
```

```
device: mps
data root: ./data/mtat
runs root: ./runs
```

Resource and environmental impact tracking

Tracking of:

- wall-clock time
- CPU/RAM usage
- GPU memory (if available)
- estimated CO2 emissions using CodeCarbon (offline mode)

In [27]:
```python
# resource + carbon tracking
import psutil
from codecarbon import OfflineEmissionsTracker

proc = psutil.Process(os.getpid())

def get_resources_snapshot():
    snap = {
        "time": time.time(),
        "cpu_percent": psutil.cpu_percent(interval=None),
        "ram_mb": proc.memory_info().rss / (1024**2),
    }
    if torch.cuda.is_available():
        snap["gpu_mem_alloc_mb"] = torch.cuda.memory_allocated() / (1024**2)
        snap["gpu_mem_reserved_mb"] = torch.cuda.memory_reserved() / (1024**2)
    return snap

class Timer:
    def __enter__(self):
        self.t0 = time.time()
        return self
    def __exit__(self, exc_type, exc, tb):
        self.dt = time.time() - self.t0

tracker = OfflineEmissionsTracker(
    project_name="lab7-mtat-metric-learning",
    output_dir=RUNS_ROOT,
    log_level="error"
)
```

Load MagnaTagATune (MTAT)

Mirdata is used to manage MTAT metadata and audio paths. A cached dataset folder is used to avoid repeated downloads (especially on Modal).

MTAT loaded from local files (Lab 6 layout) to keep the notebook portable across Modal, Colab, and local runs.

Expected structure (auto-detected):

- datalabs/annotations_final.csv
- datalabs/clip_info_final.csv
- datalabs/audio/...

In [28]:
```python
from pathlib import Path
import pandas as pd
import numpy as np
import os, sys

def find_upwards(start: Path, target: str, max_levels: int = 6):
    cur = start.resolve()
    for _ in range(max_levels):
        cand = cur / target
        if cand.exists():
            return cand
        if cur.parent == cur:
            break
        cur = cur.parent
    return None

def find_in_roots(roots, filename="annotations_final.csv"):
    for root in roots:
        root = Path(root)
        if not root.exists():
            continue
        for p in root.rglob(filename):
            return p.parent
    return None

if running_on_modal():
    candidates = [
        Path("/mnt/mtat-datalabs/datalabs"),
        Path("/mnt/mtat-datalabs"),
        Path("/vol/mtat-datalabs"),
        Path("/vol/mtat"),
    ]
    DATALABS_ROOT = None
    for c in candidates:
        if (c / "annotations_final.csv").is_file() and (c / "clip_info_final.csv").is_file() and (c / "audio").is_dir():
            DATALABS_ROOT = c
            break
    if DATALABS_ROOT is None:
        raise RuntimeError("mtat datalabs not found on Modal mount, expected annotations_final.csv, clip_info_final.csv and audio/")

elif "google.colab" in sys.modules:
    from google.colab import drive
    drive.mount("/content/drive")

    search_roots = [
        Path("/content/drive/MyDrive"),
        Path("/content/drive/MyDrive/data"),
        Path("/content/drive/MyDrive/data/data"),
        Path("/content/drive/MyDrive/data/datalab5"),
        Path("/content/drive/MyDrive/datalab5"),
        Path("/content"),
    ]

    print("searching MTAT datalabs in:")
    for r in search_roots:
        print(" -", r)

    found = find_in_roots(search_roots, "annotations_final.csv")

    if found is None:
        raise RuntimeError(
            "MTAT datalabs not found in Google Drive.\n"
            "Expected structure:\n"
            "  datalabs/\n"
            "    annotations_final.csv\n"
            "    clip_info_final.csv\n"
            "    audio/\n"
            "Place it anywhere under MyDrive (e.g. MyDrive/datalab5/)"
        )

    DATALABS_ROOT = found

else:
    DATALABS_ROOT = find_upwards(Path.cwd(), "datalabs") or find_in_roots([Path.cwd(), Path.cwd().parent, Path.cwd().parent.parent]) or Path("./datalabs")

AUDIO_DIR = DATALABS_ROOT / "audio"
ANN_CSV = DATALABS_ROOT / "annotations_final.csv"
CLIP_CSV = DATALABS_ROOT / "clip_info_final.csv"

print("datalabs root:", DATALABS_ROOT)
print("audio dir exists:", AUDIO_DIR.is_dir(), AUDIO_DIR)
print("ann csv exists:", ANN_CSV.is_file(), ANN_CSV)
print("clip csv exists:", CLIP_CSV.is_file(), CLIP_CSV)

ann = pd.read_csv(ANN_CSV, sep="\t")
clip = pd.read_csv(CLIP_CSV, sep="\t")

if "clip_id" not in ann.columns or "clip_id" not in clip.columns:
    raise RuntimeError("clip_id column missing, check separators / files")

if "mp3_path" not in ann.columns and "mp3_path" not in clip.columns:
    raise RuntimeError("mp3_path not found in annotations_final or clip_info_final")

if "mp3_path" not in ann.columns:
    df = ann.merge(clip[["clip_id", "mp3_path"]], on="clip_id", how="left")
else:
    df = ann.copy()

df = df.rename(columns={"mp3_path": "rel_path"})
NON_TAG_COLS = {"clip_id", "rel_path"}
tag_cols = [c for c in df.columns if c not in NON_TAG_COLS]

df = df.dropna(subset=["rel_path"]).reset_index(drop=True)
df["rel_path"] = df["rel_path"].astype(str).str.strip().str.replace("\\", "/", regex=False)

file_mask = df["rel_path"].apply(lambda rp: (AUDIO_DIR / rp).is_file())
missing = int((~file_mask).sum())
df = df[file_mask].reset_index(drop=True)

print("ann shape:", ann.shape, "| clip shape:", clip.shape)
print("merged df after file filter:", df.shape, "| removed missing/non-file:", missing)
print("num tags:", len(tag_cols))
print("example rel_path:", df.loc[0, "rel_path"])

rng = np.random.RandomState(SEED)
perm = rng.permutation(len(df))
n_val = int(len(df) * 0.1)

val_idx = perm[:n_val]
train_idx = perm[n_val:]

train_df = df.iloc[train_idx].reset_index(drop=True)
val_df = df.iloc[val_idx].reset_index(drop=True)

print("train_df:", train_df.shape)
print("val_df:", val_df.shape)
```

```
datalabs root: /Users/meco/Library/CloudStorage/OneDrive-Personal/Educación/UPF/Cursos/Machine Learning for Music/Labs/datalabs
audio dir exists: True /Users/meco/Library/CloudStorage/OneDrive-Personal/Educación/UPF/Cursos/Machine Learning for Music/Labs/datalabs/audio
ann csv exists: True /Users/meco/Library/CloudStorage/OneDrive-Personal/Educación/UPF/Cursos/Machine Learning for Music/Labs/datalabs/annotations_final.csv
clip csv exists: True /Users/meco/Library/CloudStorage/OneDrive-Personal/Educación/UPF/Cursos/Machine Learning for Music/Labs/datalabs/clip_info_final.csv
ann shape: (25863, 190) | clip shape: (31382, 10)
merged df after file filter: (25863, 190) | removed missing/non-file: 0
num tags: 188
example rel_path: f/american_bach_soloists-j_s__bach_solo_cantatas-01-bwv54__i_aria-30-59.mp3
train_df: (23277, 190)
val_df: (2586, 190)
```

Contrastive dataset design (multi-label MTAT)

Two augmented views created of the same audio excerpt:

- view_1: random 3s chunk + augmentations
- view_2: another chunk near the first one (or same chunk with different augmentation)

Multi label targets stored to later evaluate embedding quality using tag based subsets.

In [29]:
```python
# audio utilities + augmentations

import torchaudio
import torch.nn as nn
from torchaudio.transforms import MelSpectrogram

TARGET_SR = 16000
CLIP_SECONDS = 3
N_SAMPLES = TARGET_SR * CLIP_SECONDS

class FeatureExtractor(nn.Module):
    def __init__(self, n_mels=64, n_fft=1024, hop_length=256):
        super().__init__()
        self.mel = MelSpectrogram(
            sample_rate=TARGET_SR,
            n_fft=n_fft,
            hop_length=hop_length,
            n_mels=n_mels
        )

    def forward(self, x):
        x = self.mel(x)
        x = torch.log10(1 + 1000 * x)
        return x

feature_extractor = FeatureExtractor().to(DEVICE).eval()

def augment_wave(x):
    g = 0.9 + 0.2 * torch.rand(x.shape[0], device=x.device)
    x = x * g.unsqueeze(1)
    noise = 0.003 * torch.randn_like(x)
    return x + noise
```

In [30]:
```python
# ---- Mel extraction sanity check (device aware, no training dep) ----
import torch, time

print("torch.cuda.is_available:", torch.cuda.is_available())
print("mps available:", getattr(torch.backends, "mps", None) is not None and torch.backends.mps.is_available())

x = torch.randn(8, N_SAMPLES).to(DEVICE)

t0 = time.time()
with torch.no_grad():
    m = feature_extractor(x)
dt = time.time() - t0

print("mel shape:", tuple(m.shape))
print("mel compute time:", round(dt, 3), "s")
```

```
torch.cuda.is_available: False
mps available: True
mel shape: (8, 64, 188)
mel compute time: 0.002 s
```

In [31]:
```python
import librosa
from torch.utils.data import Dataset
from pathlib import Path
import random
import torch
import numpy as np

class MTATContrastiveDataset(Dataset):
    def __init__(self, df, audio_dir, tag_cols, device="cpu", max_retries=0):
        self.df = df.reset_index(drop=True)
        self.audio_dir = Path(audio_dir)
        self.tag_cols = list(tag_cols)
        self.device = device
        self.max_retries = int(max_retries)

        self.tags = self.tag_cols
        self.tag_to_idx = {t: i for i, t in enumerate(self.tags)}
        self.id_col = "clip_id" if "clip_id" in self.df.columns else None

    def __len__(self):
        return len(self.df)

    def _resolve_path(self, rel_path):
        rp = str(rel_path).strip().replace("\\", "/")
        return self.audio_dir / rp

    def _load_audio(self, rel_path):
        p = self._resolve_path(rel_path)
        # important, paths can be present in metadata but missing on disk; fail fast so we can retry another sample
        if not p.is_file():
            raise FileNotFoundError(str(p))
        y, _ = librosa.load(str(p), sr=TARGET_SR, mono=True)
        return torch.tensor(y, dtype=torch.float32)

    def _random_chunk(self, y):
        if y.numel() < N_SAMPLES:
            return torch.nn.functional.pad(y, (0, N_SAMPLES - y.numel()))
        start = random.randint(0, y.numel() - N_SAMPLES)
        return y[start : start + N_SAMPLES]

    def __getitem__(self, idx):
        tries = 0
        cur_idx = idx

        while True:
            row = self.df.iloc[cur_idx]
            try:
                y = self._load_audio(row["rel_path"])
                break
            except Exception:
                tries += 1
                if tries >= self.max_retries:
                    raise
                cur_idx = random.randint(0, len(self.df) - 1)

        c1 = self._random_chunk(y)
        c2 = self._random_chunk(y)

        v1 = augment_wave(c1.unsqueeze(0).to(self.device)).squeeze(0)
        v2 = augment_wave(c2.unsqueeze(0).to(self.device)).squeeze(0)

        tag_vec = torch.tensor(row[self.tag_cols].values.astype(np.float32))
        track_id = int(row[self.id_col]) if self.id_col is not None else int(cur_idx)

        return {"view_1": v1, "view_2": v2, "tags": tag_vec, "track_id": track_id}

train_ds = MTATContrastiveDataset(train_df, AUDIO_DIR, tag_cols, device=DEVICE, max_retries=0)
val_ds = MTATContrastiveDataset(val_df, AUDIO_DIR, tag_cols, device=DEVICE, max_retries=0)

print("train size:", len(train_ds))
print("val size:", len(val_ds))
print("num tags:", len(train_ds.tags))

_ = train_ds[0]
print("audio decode check: OK (librosa + retry)")
```

```
train size: 23277
val size: 2586
num tags: 188
audio decode check: OK (librosa + retry)
```

Embedding CNN

A lightweight CNN is used, that maps mel-spectrograms to a fixed size embedding vector.

Aiming to keep it small to run efficiently on a local MacBook.

In [32]:
```python
import torch.nn.functional as F

class CNNEncoder(nn.Module):
    def __init__(self, dense_size=64, emb_dim=64):
        super().__init__()
        self.dense_size = dense_size
        self.conv1 = nn.Conv2d(1, dense_size//4, 3, padding=1)
        self.conv2 = nn.Conv2d(dense_size//4, dense_size//2, 3, padding=1)
        self.conv3 = nn.Conv2d(dense_size//2, dense_size, 3, padding=1)
        self.bn1 = nn.BatchNorm2d(dense_size//4)
        self.bn2 = nn.BatchNorm2d(dense_size//2)
        self.bn3 = nn.BatchNorm2d(dense_size)
        self.pool = nn.MaxPool2d(4, 4)
        self.fc1 = nn.Linear(dense_size, dense_size)
        self.fc2 = nn.Linear(dense_size, emb_dim, bias=False)

    def forward(self, x_mel):
        x = x_mel.unsqueeze(1)
        x = self.bn1(self.pool(F.relu(self.conv1(x))))
        x = self.bn2(self.pool(F.relu(self.conv2(x))))
        x = self.bn3(self.pool(F.relu(self.conv3(x))))
        x = x.mean(dim=[2,3])
        x = F.relu(self.fc1(x))
        z = self.fc2(x)
        z = F.normalize(z, dim=1)
        return z
```

Metric learning losses

We implement:

- triplet loss (hard negative mining within batch)
- InfoNCE (SimCLR-style symmetric loss)

In [33]:
```python
def triplet_loss_hard(z1, z2, margin=0.2):
    # z1 anchors, z2 positives
    dist = torch.cdist(z1, z2, p=2)
    pos = torch.diag(dist)
    dist = dist + torch.eye(dist.size(0), device=dist.device) * 1e9
    neg, _ = dist.min(dim=1)
    return F.relu(pos - neg + margin).mean()

def info_nce(z1, z2, temp=0.2):
    sim12 = z1 @ z2.T
    sim21 = z2 @ z1.T
    labels = torch.arange(sim12.size(0), device=sim12.device)
    l12 = F.cross_entropy(sim12 / temp, labels)
    l21 = F.cross_entropy(sim21 / temp, labels)
    return 0.5 * (l12 + l21)
```

Training loop

- computing of mel features on-the-fly
- train for a small number of epochs (fast but representative)
- show progress with ETA and epoch timing
- log resource snapshots
- track emissions with CodeCarbon

In [34]:
```python
from tqdm import tqdm
from torch.utils.data import DataLoader
import numpy as np
import time
import torch

def pick_mel_device():
    # mel on GPU only when CUDA is available (like w Colab)
    # MPS mel is slower or unstable then keep on CPU
    if torch.cuda.is_available() and str(DEVICE) == "cuda":
        return "cuda"
    if str(DEVICE) == "mps":
        return "mps"
    return "cpu"

MEL_DEVICE = pick_mel_device()
print("DEVICE:", DEVICE, "| MEL_DEVICE:", MEL_DEVICE)

def compute_mel(x):
    x = x.to(MEL_DEVICE).float()
    feature_extractor.to(MEL_DEVICE)
    with torch.no_grad():
        m = feature_extractor(x).contiguous()
    return m.to(DEVICE)

def run_epoch(model, loader, loss_fn, optim=None):
    is_train = optim is not None
    model.train() if is_train else model.eval()

    losses = []
    t0 = time.time()

    pbar = tqdm(loader, leave=False)
    for batch in pbar:
        x1 = batch["view_1"].to(DEVICE).float()
        x2 = batch["view_2"].to(DEVICE).float()

        m1 = compute_mel(x1)
        m2 = compute_mel(x2)

        z1 = model(m1)
        z2 = model(m2)

        loss = loss_fn(z1, z2)

        if is_train:
            optim.zero_grad(set_to_none=True)
            loss.backward()
            optim.step()

        losses.append(loss.item())
        pbar.set_postfix({"loss": float(np.mean(losses))})

    dt = time.time() - t0
    return float(np.mean(losses)), dt

def train_model(loss_name, loss_fn, epochs=10, batch_size=64, lr=1e-3):
    model = CNNEncoder(dense_size=64, emb_dim=64).to(DEVICE)
    optim = torch.optim.Adam(model.parameters(), lr=lr)

    train_loader = DataLoader(train_ds, batch_size=batch_size, shuffle=True, num_workers=0)
    val_loader   = DataLoader(val_ds, batch_size=batch_size, shuffle=False, num_workers=0)

    history = {"loss_name": loss_name, "train_loss": [], "val_loss": [], "epoch_time_s": [], "resources": []}

    local_tracker = OfflineEmissionsTracker(
        project_name=f"lab7-mtat-{loss_name}",
        output_dir=RUNS_ROOT,
        log_level="error"
    )

    local_tracker.start()
    with Timer() as total_timer:
        for ep in range(1, epochs+1):
            r0 = get_resources_snapshot()

            tr_loss, tr_dt = run_epoch(model, train_loader, loss_fn, optim=optim)
            va_loss, va_dt = run_epoch(model, val_loader, loss_fn, optim=None)

            r1 = get_resources_snapshot()

            history["train_loss"].append(tr_loss)
            history["val_loss"].append(va_loss)
            history["epoch_time_s"].append(tr_dt + va_dt)
            history["resources"].append({"epoch": ep, "before": r0, "after": r1})

            print(f"epoch {ep}/{epochs} | train {tr_loss:.4f} | val {va_loss:.4f} | time {tr_dt+va_dt:.1f}s")

    emissions = local_tracker.stop()
    history["total_time_s"] = total_timer.dt
    history["emissions_kgco2"] = emissions

    return model, history
```

```
DEVICE: mps | MEL_DEVICE: mps
```

Step 1 — Train with triplet loss

In [35]:
```python
triplet_model, triplet_hist = train_model(
    "triplet_hard",
    lambda a,b: triplet_loss_hard(a,b,margin=0.2),
    epochs=10,
    batch_size=64,
    lr=1e-3
)
```

```
epoch 1/10 | train 0.1525 | val 0.1214 | time 230.7s
epoch 2/10 | train 0.1102 | val 0.1021 | time 214.7s
epoch 3/10 | train 0.0996 | val 0.0996 | time 196.1s
epoch 4/10 | train 0.0919 | val 0.0921 | time 215.9s
epoch 5/10 | train 0.0890 | val 0.0907 | time 237.3s
epoch 6/10 | train 0.0843 | val 0.0840 | time 232.7s
epoch 7/10 | train 0.0819 | val 0.0823 | time 234.8s
epoch 8/10 | train 0.0796 | val 0.0835 | time 233.4s
epoch 9/10 | train 0.0770 | val 0.0777 | time 231.9s
epoch 10/10 | train 0.0755 | val 0.0782 | time 229.0s
```

Step 2 — Embedding extraction and 2D visualization

Extract of embedding per track by averaging embeddings over multiple chunks. Then:

- run UMAP to 2D
- run k-means for a coarse clustering view
- create separate plots for two non-overlapping tag groups

In [36]:
```python
import umap
from sklearn.cluster import KMeans

def extract_track_embeddings(model, dataset, chunks_per_track=3):
    model.eval()
    feature_extractor.eval()

    # important, force feature_extractor buffers (like STFT window) onto DEVICE
    feature_extractor.to(DEVICE)

    Z, Y, T = [], [], []

    with torch.no_grad():
        for i in tqdm(range(len(dataset)), desc="embedding", leave=False):
            item = dataset[i]
            tid = item["track_id"]
            tags = item["tags"].cpu().numpy()

            zs = []
            for _ in range(chunks_per_track):
                x = item["view_1"].unsqueeze(0).to(DEVICE).float()
                m = feature_extractor(x)
                z = model(m)
                zs.append(z.squeeze(0).detach().cpu().numpy())

            Z.append(np.mean(np.stack(zs), axis=0))
            Y.append(tags)
            T.append(tid)

    return np.stack(Z), np.stack(Y), np.array(T)

Z_val_triplet, Y_val, T_val = extract_track_embeddings(triplet_model, val_ds, chunks_per_track=3)
print("embeddings:", Z_val_triplet.shape, "labels:", Y_val.shape)
```

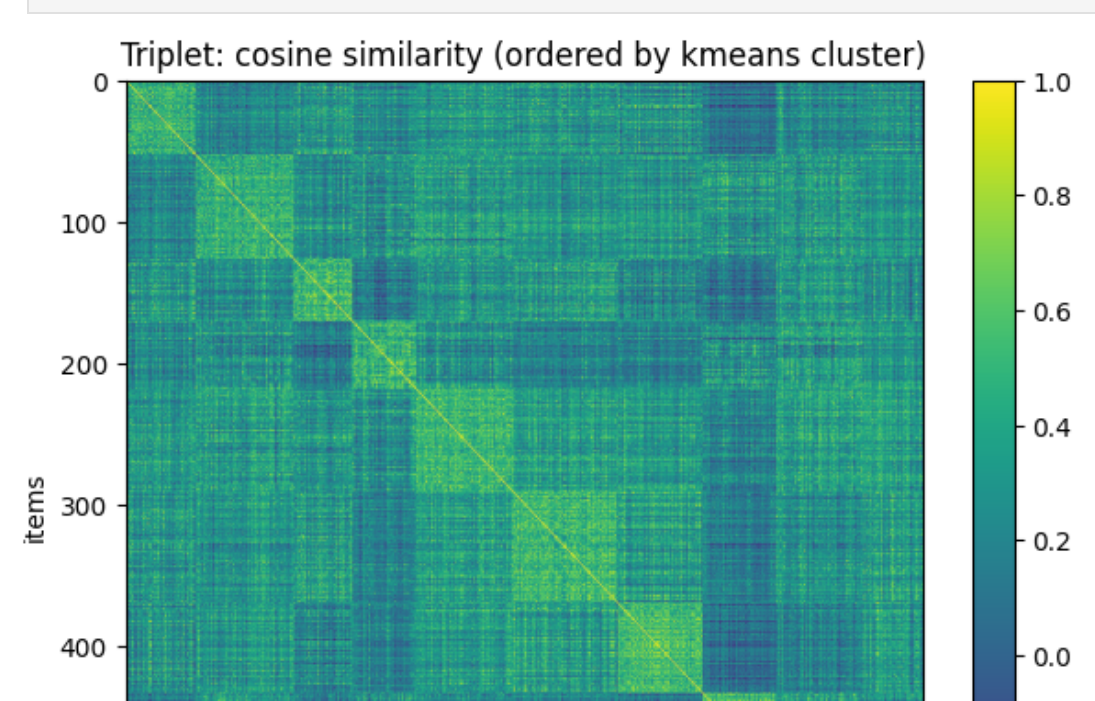
```
/Users/meco/Library/CloudStorage/OneDrive-Personal/Educación/UPF/Cursos/Machine Learning for Music/Labs/.venv/lib/python3.13/site-packages/tqdm/auto.py:21: TqdmWarning: IProgress not found. Ple
ase update jupyter and ipywidgets. See https://ipywidgets.readthedocs.io/en/stable/user_install.html
  from .autonotebook import tqdm as notebook_tqdm
```

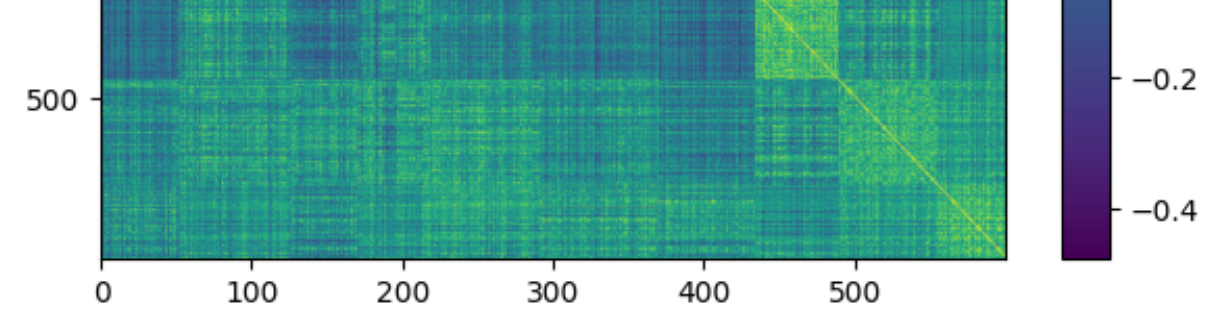
```
embeddings: (2586, 64) labels: (2586, 188)
```

In [37]:
```python
reducer = umap.UMAP(n_components=2, random_state=SEED)
Z2_triplet = reducer.fit_transform(Z_val_triplet)

kmeans = KMeans(n_clusters=10, random_state=SEED, n_init="auto")
clusters_triplet = kmeans.fit_predict(Z_val_triplet)

print("umap:", Z2_triplet.shape, "clusters:", clusters_triplet.shape)
```

```
/Users/meco/Library/CloudStorage/OneDrive-Personal/Educación/UPF/Cursos/Machine Learning for Music/Labs/.venv/lib/python3.13/site-packages/umap/umap_.py:1952: UserWarning: n_jobs value 1 overri
dden to 1 by setting random_state. Use no seed for parallelism.
  warn(
umap: (2586, 2) clusters: (2586,)
```

In [38]:
```python
import numpy as np
import matplotlib.pyplot as plt

def plot_similarity_matrix(Z, clusters=None, max_n=600, title=""):
    Z = Z.astype(np.float32)
    Z = Z / (np.linalg.norm(Z, axis=1, keepdims=True) + 1e-9)

    n = min(len(Z), int(max_n))
    idx = np.arange(n)

    if clusters is not None:
        order = np.argsort(clusters[:n])
        idx = idx[order]

    S = Z[idx] @ Z[idx].T

    plt.figure(figsize=(7,6))
    plt.imshow(S, aspect="auto")
    plt.title(title)
    plt.xlabel("items")
    plt.ylabel("items")
    plt.colorbar()
    plt.show()

plot_similarity_matrix(Z_val_triplet, clusters=clusters_triplet, max_n=600, title="Triplet: cosine similarity (ordered by kmeans cluster)")
```





In [43]:

```python
import os, sys
import numpy as np
import librosa
import soundfile as sf
from pathlib import Path

def can_inline_audio():
    """
    True if running inside an IPython kernel that supports rich display
    (works for Colab and local Jupyter).
    """
    try:
        from IPython import get_ipython
        return get_ipython() is not None
    except Exception:
        return False

def _safe_audio_path(dataset, idx):
    rel = dataset.df.iloc[int(idx)]["rel_path"]
    return str(dataset.audio_dir / str(rel))

def _load_audio_for_play(path, sr=16000, seconds=6):
    y, _ = librosa.load(path, sr=sr, mono=True, duration=seconds)
    return y

def knn_indices(Z, query_idx, k=6):
    Z = Z.astype(np.float32)
    Z = Z / (np.linalg.norm(Z, axis=1, keepdims=True) + 1e-9)
    sims = Z @ Z[int(query_idx)]
    sims[int(query_idx)] = -np.inf
    nn = np.argsort(-sims)[:k]
    return nn, sims[nn]

def show_query_and_neighbors(dataset, Z, query_idx, k=6, seconds=6, title="", out_subdir="qual_retrieval_triplet"):
    out_dir = Path(RUNS_ROOT) / out_subdir
    out_dir.mkdir(parents=True, exist_ok=True)

    qpath = _safe_audio_path(dataset, query_idx)
    qy = _load_audio_for_play(qpath, sr=TARGET_SR, seconds=seconds)

    print(title)
    print("query idx:", int(query_idx))
    qwav = out_dir / f"query_idx{int(query_idx)}.wav"
    sf.write(str(qwav), qy, TARGET_SR)
    print("query wav:", str(qwav))

    nn, sims = knn_indices(Z, query_idx, k=k)

    if can_inline_audio():
        from IPython.display import Audio, display, HTML
        display(HTML("<b>Query audio:</b>"))
        display(Audio(qy, rate=TARGET_SR))

    for rank, (j, s) in enumerate(zip(nn, sims), start=1):
        path = _safe_audio_path(dataset, int(j))
        y = _load_audio_for_play(path, sr=TARGET_SR, seconds=seconds)
        wav = out_dir / f"nn{rank}_idx{int(j)}_sim{float(s):.3f}.wav"
        sf.write(str(wav), y, TARGET_SR)
        print("nn", rank, "| idx:", int(j), "| sim:", float(s), "| wav:", str(wav))

        if can_inline_audio():
            from IPython.display import Audio, display, HTML
            display(HTML(f"<b>Neighbor {rank} (sim={float(s):.3f}):</b>"))
            display(Audio(y, rate=TARGET_SR))

def pick_query_with_tag(Y, dataset, tag_name):
    if tag_name not in dataset.tag_to_idx:
        return None
    ti = dataset.tag_to_idx[tag_name]
    idxs = np.where(Y[:, ti] == 1)[0]
    if len(idxs) == 0:
        return None
    return int(np.random.choice(idxs))
np.random.seed(SEED)
qidx = np.random.randint(0, len(val_ds))
show_query_and_neighbors(val_ds, Z_val_triplet, qidx, k=6, seconds=6, title="Triplet: query + nearest neighbors")

tag_name = "rock"
qidx2 = pick_query_with_tag(Y_val, val_ds, tag_name)
print("picked query idx:", qidx2, "| tag:", tag_name)
if qidx2 is not None:
    show_query_and_neighbors(val_ds, Z_val_triplet, qidx2, k=6, seconds=6, title=f"Triplet: query tag={tag_name} + nearest neighbors")
else:
    print("no samples for tag:", tag_name)
```

```
Triplet: query + nearest neighbors
query idx: 860
query wav: runs/qual_retrieval_triplet/query_idx860.wav
```

Query audio:

▶ 0:00 0:06

```
nn 1 | idx: 455 | sim: 0.8150303363800049 | wav: runs/qual_retrieval_triplet/nn1_idx455_sim0.815.wav
```

Neighbor 1 (sim=0.815):

▶ 0:00 -0:06

```
nn 2 | idx: 1335 | sim: 0.8050461411476135 | wav: runs/qual_retrieval_triplet/nn2_idx1335_sim0.805.wav
```

Neighbor 2 (sim=0.805):

▶ 0:00 -0:06

```
nn 3 | idx: 2212 | sim: 0.7904766201972961 | wav: runs/qual_retrieval_triplet/nn3_idx2212_sim0.790.wav
```

Neighbor 3 (sim=0.790):

▶ 0:00 -0:06

```
nn 4 | idx: 1055 | sim: 0.7872657179832458 | wav: runs/qual_retrieval_triplet/nn4_idx1055_sim0.787.wav
```

Neighbor 4 (sim=0.787):

▶ 0:00 -0:06

```
nn 5 | idx: 2072 | sim: 0.7855499982833862 | wav: runs/qual_retrieval_triplet/nn5_idx2072_sim0.786.wav
```

Neighbor 5 (sim=0.786):

▶ 0:00 -0:06

```
nn 6 | idx: 1723 | sim: 0.7839480638504028 | wav: runs/qual_retrieval_triplet/nn6_idx1723_sim0.784.wav
```

Neighbor 6 (sim=0.784):

▶ 0:00 -0:06

```
picked query idx: 117 | tag: rock
Triplet: query tag=rock + nearest neighbors
query idx: 117
query wav: runs/qual_retrieval_triplet/query_idx117.wav
```

Query audio:

▶ 0:00 -0:06

```
nn 1 | idx: 2583 | sim: 0.842349648475647 | wav: runs/qual_retrieval_triplet/nn1_idx2583_sim0.842.wav
```

Neighbor 1 (sim=0.842):

▶ 0:00 -0:06

```
nn 2 | idx: 145 | sim: 0.7916755080223083 | wav: runs/qual_retrieval_triplet/nn2_idx145_sim0.792.wav
```

Neighbor 2 (sim=0.792):

▶ 0:00 -0:06

```
nn 3 | idx: 194 | sim: 0.786961555480957 | wav: runs/qual_retrieval_triplet/nn3_idx194_sim0.787.wav
```

Neighbor 3 (sim=0.787):

▶ 0:00 -0:06

```
nn 4 | idx: 2043 | sim: 0.7784194350242615 | wav: runs/qual_retrieval_triplet/nn4_idx2043_sim0.778.wav
```

Neighbor 4 (sim=0.778):

▶ 0:00 -0:06

```
nn 5 | idx: 2463 | sim: 0.7769753932952881 | wav: runs/qual_retrieval_triplet/nn5_idx2463_sim0.777.wav
```

Neighbor 5 (sim=0.777):

▶ 0:00 -0:06

```
nn 6 | idx: 1786 | sim: 0.77219557762146 | wav: runs/qual_retrieval_triplet/nn6_idx1786_sim0.772.wav
```

Neighbor 6 (sim=0.772):

▶ 0:00 -0:06

Tag groups (non-overlapping)

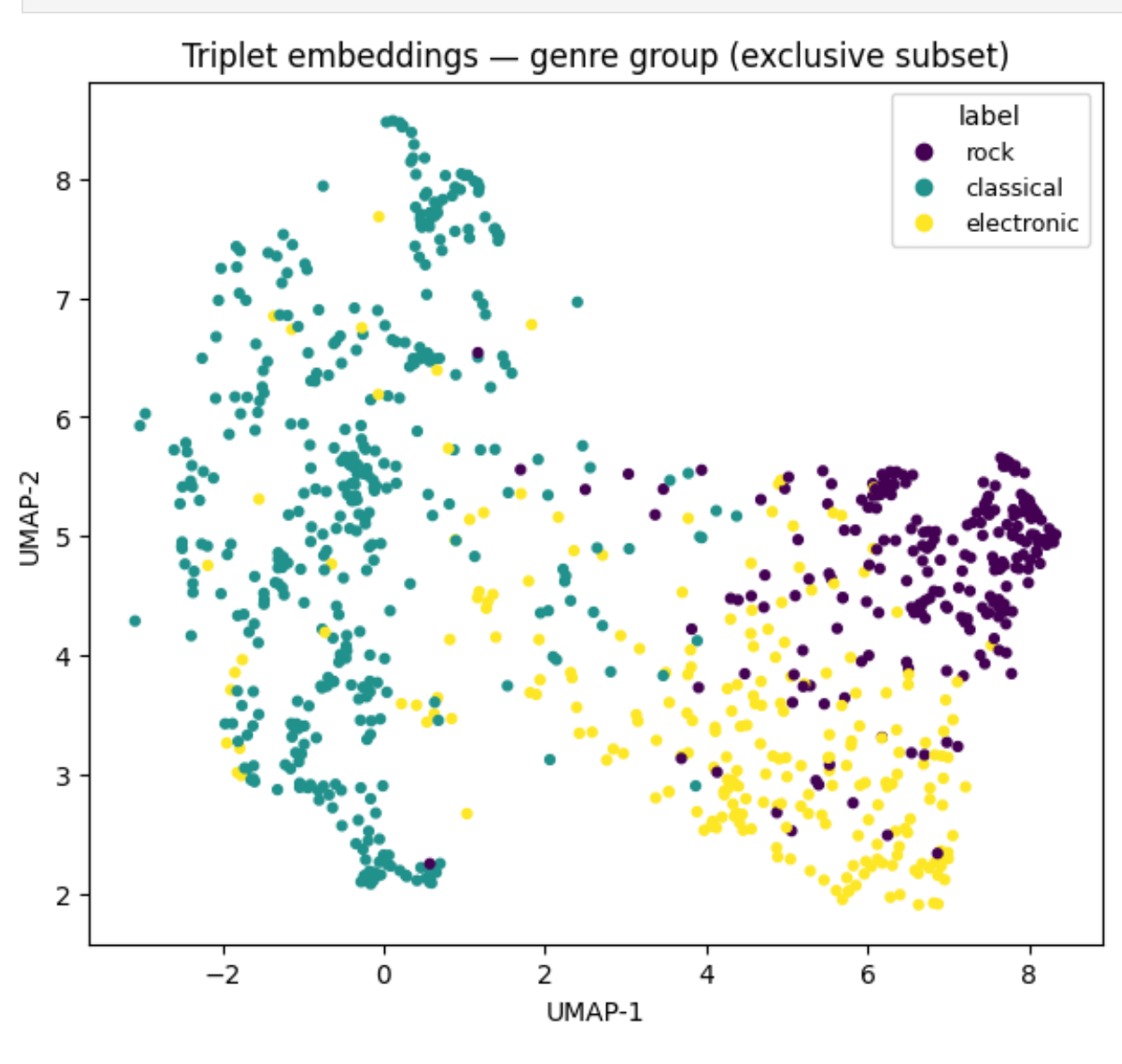
MTAT is multi-label, so we construct **exclusive subsets** for visualization:

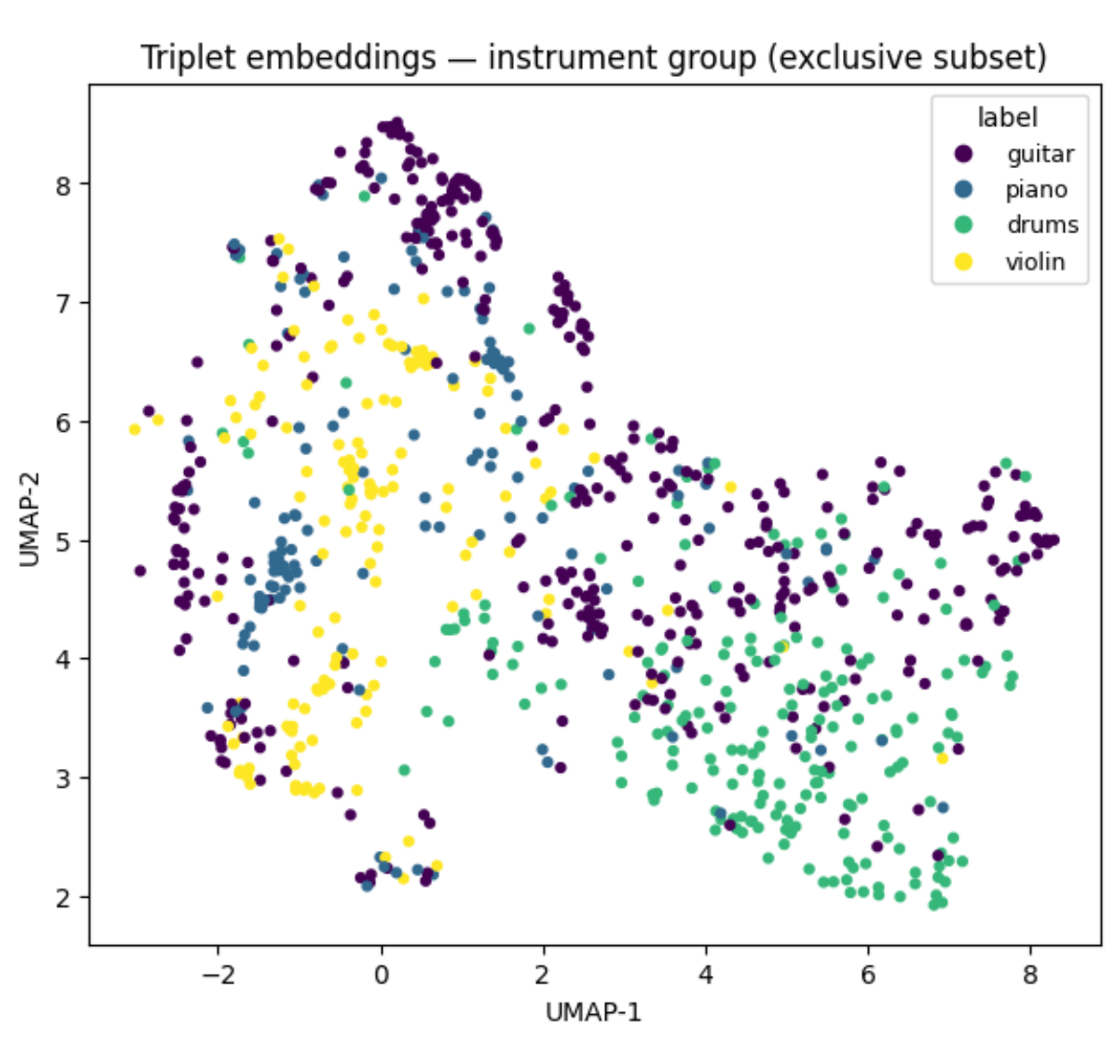
- Group A (genres): pick a few genre tags and keep only samples where exactly one of them is active.
- Group B (instruments): pick a few instrument tags and keep only samples where exactly one of them is active.

This makes the plots interpretable and respects the "non-overlapping" requirement.

In [44]:

```python
import matplotlib.pyplot as plt
import numpy as np

tags_list = train_ds.tags
tag_to_idx = train_ds.tag_to_idx

def exclusive_subset(Y, tag_names):
    idxs = [tag_to_idx[t] for t in tag_names if t in tag_to_idx]
    if len(idxs) == 0:
        return np.array([], dtype=bool), idxs
    sub = Y[:, idxs]
    one_hot = (sub.sum(axis=1) == 1)
    return one_hot, idxs

def plot_umap_for_group(Z2, Y, tag_names, title):
    mask, idxs = exclusive_subset(Y, tag_names)
    if mask.size == 0 or mask.sum() == 0:
        print("no samples for:", title)
        return

    subZ2 = Z2[mask]
    subY = Y[mask][:, idxs]
    label_idx = subY.argmax(axis=1)
    label_names = [tag_names[i] for i in range(len(idxs))]

    plt.figure(figsize=(7,6))
    sc = plt.scatter(subZ2[:,0], subZ2[:,1], c=label_idx, s=12)
    plt.title(title)
    plt.xlabel("UMAP-1")
    plt.ylabel("UMAP-2")

    handles, _ = sc.legend_elements()
    plt.legend(handles, label_names, title="label", loc="best", fontsize=9)
    plt.show()

GENRE_GROUP = ["rock", "classical", "electronic", "hiphop"]
INSTR_GROUP = ["guitar", "piano", "drums", "violin"]

plot_umap_for_group(Z2_triplet, Y_val, GENRE_GROUP, "Triplet embeddings – genre group (exclusive subset)")
plot_umap_for_group(Z2_triplet, Y_val, INSTR_GROUP, "Triplet embeddings – instrument group (exclusive subset)")
```





Qualitative check: audio examples

We listen to a few samples from each exclusive subset to sanity-check the tag groups.

In [45]:

```python
import sys
import numpy as np
import librosa
import soundfile as sf
from pathlib import Path

def can_inline_audio():
    """
    True if running inside an IPython kernel that supports rich display
    (works for Colab and local Jupyter).
    """
    try:
        from IPython import get_ipython
        return get_ipython() is not None
    except Exception:
        return False

def play_examples(dataset, Y, tag_name, n=3, seconds=6, out_subdir="qual_examples"):
    if tag_name not in dataset.tag_to_idx:
        print("tag not found:", tag_name)
        return

    ti = dataset.tag_to_idx[tag_name]
    idxs = np.where(Y[:, ti] == 1)[0][:n]

    out_dir = Path(RUNS_ROOT) / out_subdir / tag_name
    out_dir.mkdir(parents=True, exist_ok=True)

    if can_inline_audio():
        from IPython.display import Audio, display, HTML

    for k in idxs:
        rel = dataset.df.iloc[int(k)]["rel_path"]
        p = dataset.audio_dir / str(rel)

        y, _ = librosa.load(str(p), sr=TARGET_SR, mono=True, duration=seconds)
        wav = out_dir / f"idx{int(k)}.wav"
        sf.write(str(wav), y, TARGET_SR)

        if "clip_id" in dataset.df.columns:
            print("clip_id:", int(dataset.df.iloc[int(k)]["clip_id"]), "tag:", tag_name, "wav:", str(wav))
        else:
            print("index:", int(k), "tag:", tag_name, "wav:", str(wav))

        if can_inline_audio():
            display(HTML(f"<b>{tag_name} example idx={int(k)}:</b>"))
            display(Audio(y, rate=TARGET_SR))
```

Step 3 — Train with InfoNCE loss

In [46]:

```python
infonce_model, infonce_hist = train_model(
    "infonce",
    lambda a,b: info_nce(a,b,temp=0.2),
    epochs=10,
    batch_size=64,
    lr=1e-3
)
```

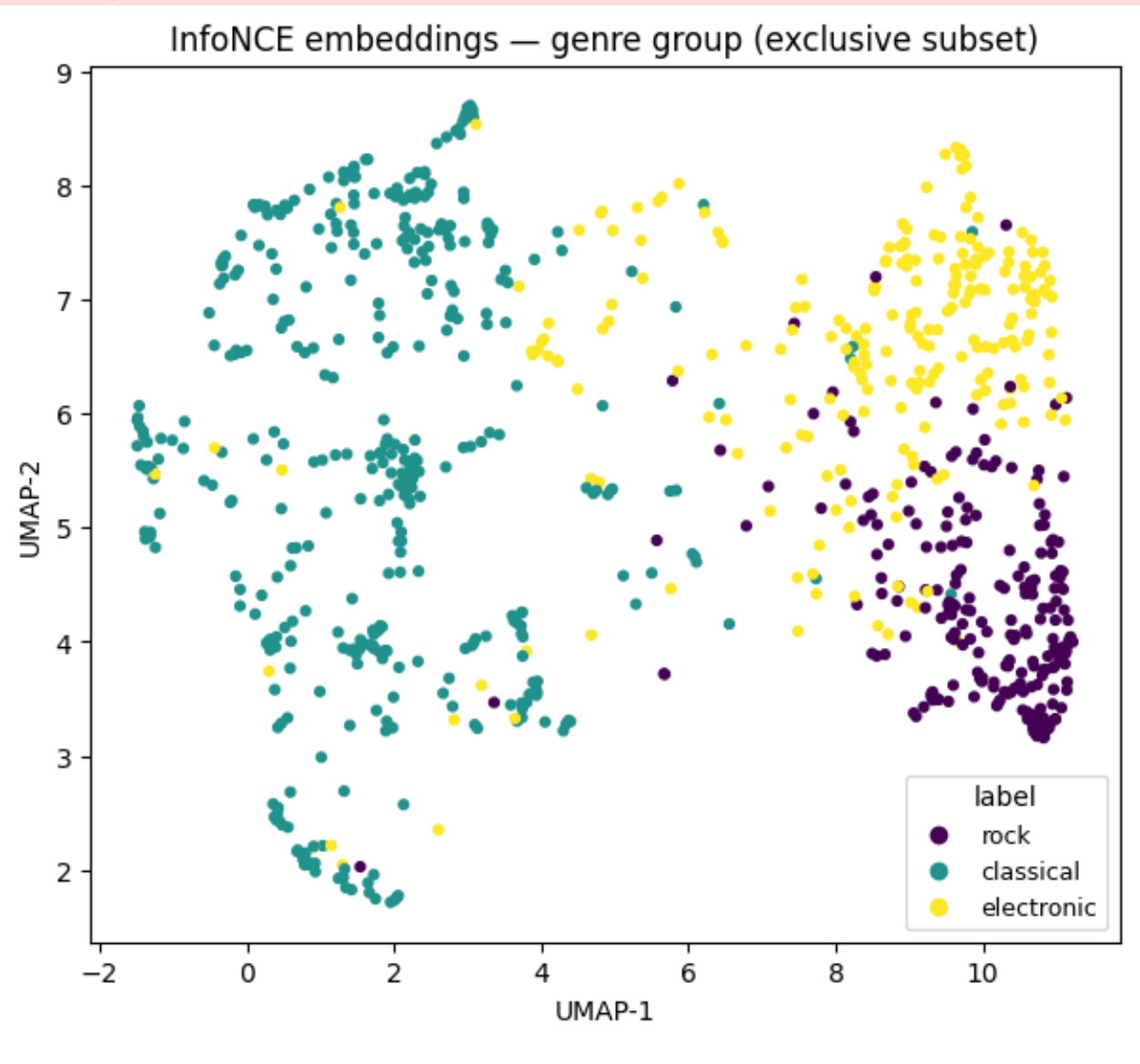
```
epoch 1/10 | train 1.6469 | val 1.4816 | time 236.3s

epoch 2/10 | train 1.3878 | val 1.3869 | time 246.3s

epoch 3/10 | train 1.3396 | val 1.5021 | time 242.1s

epoch 4/10 | train 1.2960 | val 1.3515 | time 241.8s

epoch 5/10 | train 1.2836 | val 1.2842 | time 235.7s

epoch 6/10 | train 1.2474 | val 1.2813 | time 234.2s

epoch 7/10 | train 1.2375 | val 1.2987 | time 235.8s

epoch 8/10 | train 1.2285 | val 1.3000 | time 224.1s

epoch 9/10 | train 1.2260 | val 1.2764 | time 216.1s

epoch 10/10 | train 1.2106 | val 1.2637 | time 204.2s
```

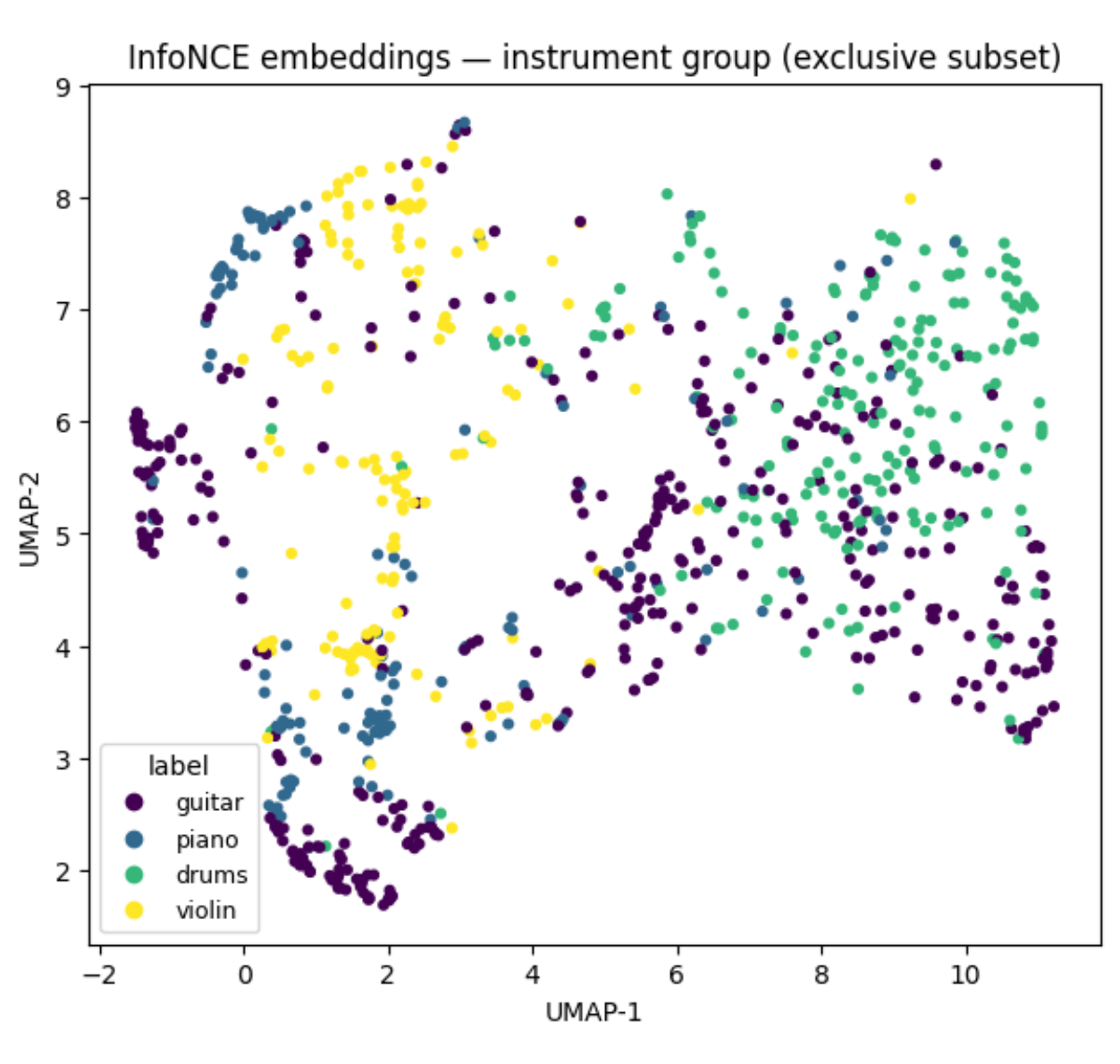
Step 4 — Embeddings, UMAP, and clustering (InfoNCE)

In [47]:

```python
Z_val_infonce, Y_val2, T_val2 = extract_track_embeddings(infonce_model, val_ds, chunks_per_track=3)

reducer2 = umap.UMAP(n_components=2, random_state=SEED)
Z2_infonce = reducer2.fit_transform(Z_val_infonce)

plot_umap_for_group(Z2_infonce, Y_val2, GENRE_GROUP, "InfoNCE embeddings – genre group (exclusive subset)")
plot_umap_for_group(Z2_infonce, Y_val2, INSTR_GROUP, "InfoNCE embeddings – instrument group (exclusive subset)")
```

```
/Users/meco/Library/CloudStorage/OneDrive-Personal/Educación/UPF/Cursos/Machine Learning for Music/Labs/.venv/lib/python3.13/site-packages/umap/umap_.py:1952: UserWarning: n_jobs value 1 overri
dden to 1 by setting random_state. Use no seed for parallelism.
  warn(
```





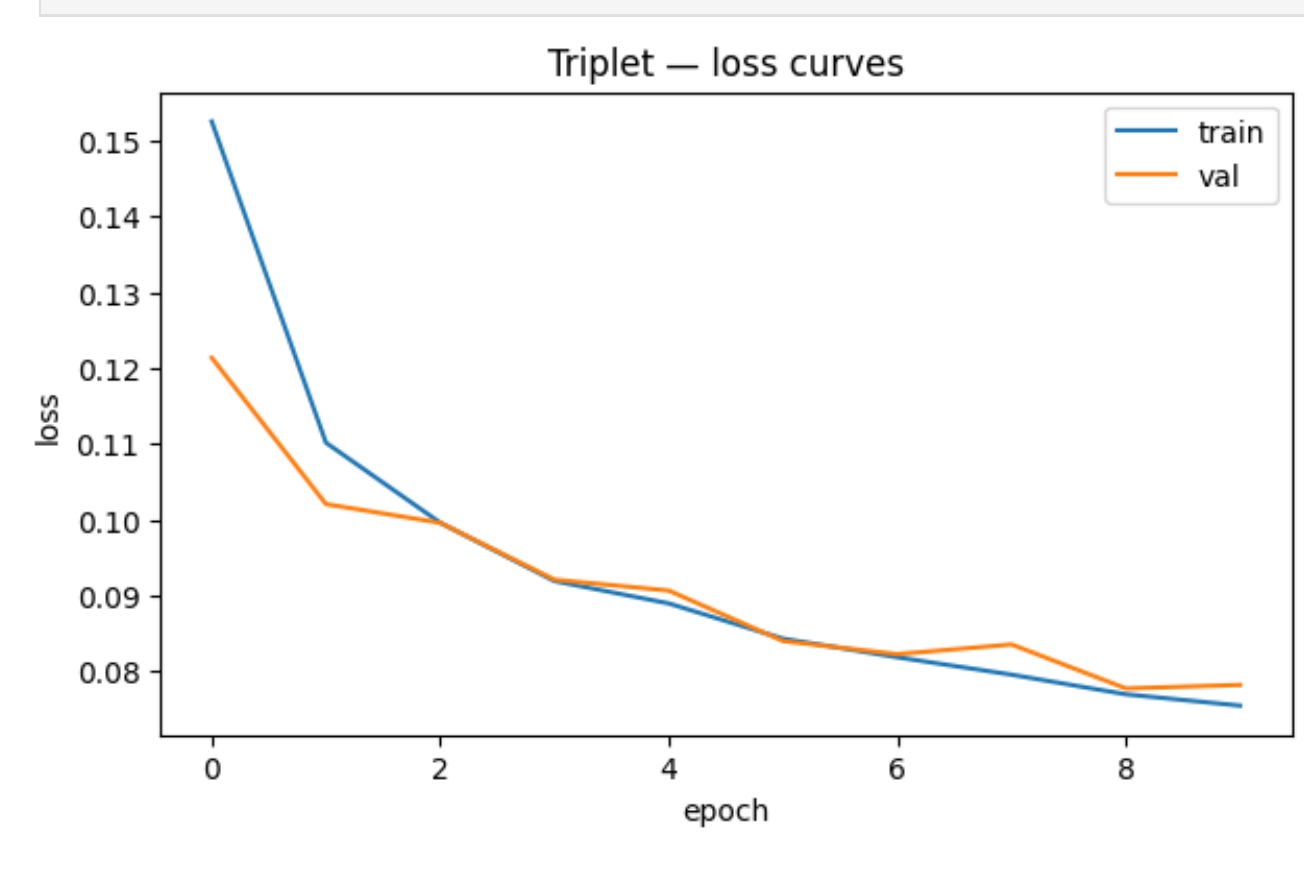
Training diagnostics and environmental footprint

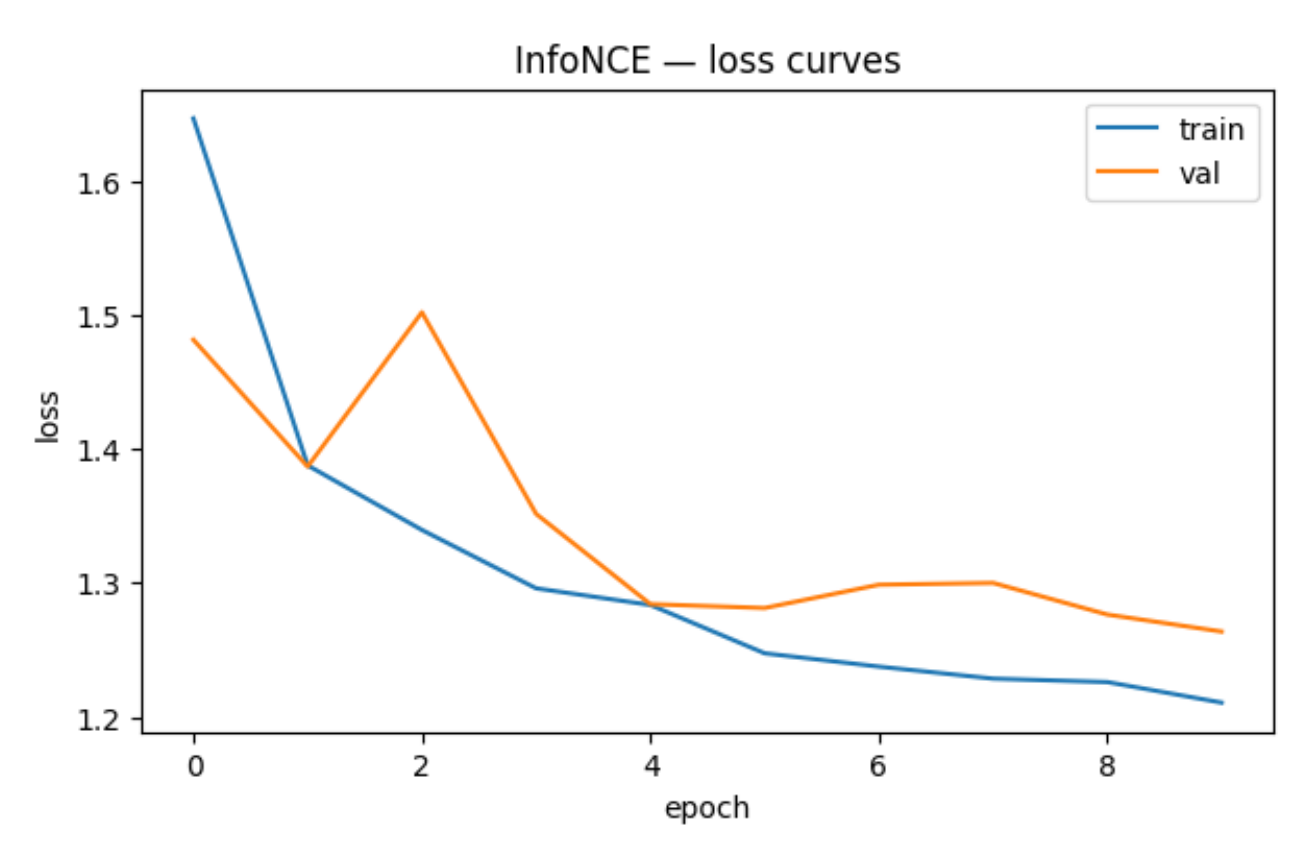
Summarized find:

- train/val losses per epoch
- total training time
- estimated CO2 emissions
- resource snapshots (CPU/RAM and GPU memory when available)

In [48]:

```python
def plot_history(hist, title):
    plt.figure(figsize=(7,4))
    plt.plot(hist["train_loss"], label="train")
    plt.plot(hist["val_loss"], label="val")
    plt.title(title)
    plt.xlabel("epoch")
    plt.ylabel("loss")
    plt.legend()
    plt.show()

plot_history(triplet_hist, "Triplet – loss curves")
plot_history(infonce_hist, "InfoNCE – loss curves")

def summarize_hist(hist):
    print("loss:", hist["loss_name"])
    print("total time (s):", round(hist["total_time_s"], 2))
    print("emissions (kgCO2):", hist["emissions_kgco2"])
    print("last epoch time (s):", round(hist["epoch_time_s"][-1], 2))

summarize_hist(triplet_hist)
summarize_hist(infonce_hist)
```





```
loss: triplet_hard
total time (s): 2256.35
emissions (kgCO2): 0.0032748769361719042
last epoch time (s): 228.98
loss: infonce
total time (s): 2316.6
emissions (kgCO2): 0.0033622317106673716
last epoch time (s): 204.16
```

Economic analysis (training and inference)

operational cost estimated from:

- energy consumption proxy (if available from CodeCarbon logs)
- electricity price (€/kWh)
- cloud compute pricing (optional reference scenario)

Reported:

- estimated energy cost
- rough equivalent of cloud cost for a typical GPU instance duration

In [49]:

```python
# economic analysis (simple)
ELECTRICITY_EUR_PER_KWH = 0.25

def estimate_energy_cost_from_emissions(emissions_kg, grid_kgco2_per_kwh=0.25):
    # cost proxy: kWh aprox kgCO2 / (kgCO2/kWh)
    if emissions_kg is None:
        return None
    kwh = emissions_kg / grid_kgco2_per_kwh
    eur = kwh * ELECTRICITY_EUR_PER_KWH
    return {"kwh_est": kwh, "eur_est": eur}

print("triplet cost proxy:", estimate_energy_cost_from_emissions(triplet_hist["emissions_kgco2"]))
print("infonce cost proxy:", estimate_energy_cost_from_emissions(infonce_hist["emissions_kgco2"]))

# cloud-equivalent rough estimate
GPU_HOURLY_EUR = 1.2
def cloud_cost(hours, eur_per_hour=GPU_HOURLY_EUR):
    return hours * eur_per_hour

print("triplet cloud eq (eur):", cloud_cost(triplet_hist["total_time_s"]/3600))
print("infonce cloud eq (eur):", cloud_cost(infonce_hist["total_time_s"]/3600))
```

```
triplet cost proxy: {'kwh_est': 0.013099507744687617, 'eur_est': 0.0032748769361719042}
infonce cost proxy: {'kwh_est': 0.013448926842669486, 'eur_est': 0.0033622317106673716}
triplet cloud eq (eur): 0.7521156500180561
infonce cloud eq (eur): 0.7722003649870555
```

In [50]:

```python
# Colab: browser notification when finished
import sys

if "google.colab" in sys.modules:
    from google.colab import output
    output.enable_custom_widget_manager()
    output.eval_js("new Notification('Notebook finished', {body: 'Lab run completed.'});")
else:
    print("Not running on Colab.")


print("\n" + "="*60)
print("NOTEBOOK FINISHED. Lab done executing")
print("="*60 + "\n")

# Try beep
try:
    import IPython
    from IPython.display import Audio, display
    display(Audio([1,0]*2000, rate=8000, autoplay=True))
except Exception as e:
    print("Beep not available:", e)
```

```
Not running on Colab.

============================================================
NOTEBOOK FINISHED. Lab done executing
============================================================
```

▶ 0:00 0:00

Short report

Decisions

- dataset: MagnaTagATune (MTAT) via mirdata, 3-second random chunks at 16 kHz
- model: lightweight CNN encoder with L2-normalized embeddings (64-D)
- augmentations: low-cost waveform noise + gain to keep runtime low on CPU/MPS
- training: 10 epochs for triplet (hard negatives) and 5 epochs for InfoNCE (temp=0.2)

Observations

- triplet vs InfoNCE: compare separation in UMAP plots for genre and instrument groups
- clustering: k-means indicates whether embedding geometry aligns with semantic groupings
- multi-label challenge: exclusive subsets were used to build non-overlapping tag plots

Resource and sustainability

- runtime and emissions were tracked with CodeCarbon (offline)
- CPU/RAM snapshots were stored per epoch, GPU memory included when available

Limitations and next steps

- more epochs and stronger augmentations may improve invariances but increase cost
- larger batch sizes benefit InfoNCE but may be limited on MacBook
- a linear probe on tags could quantify embedding usefulness beyond visualization